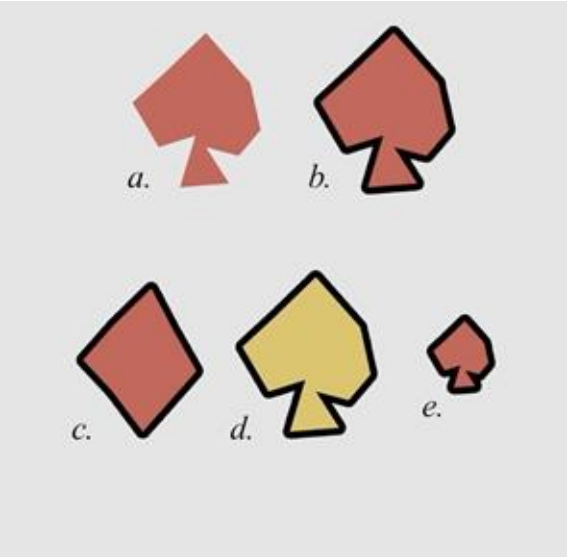

CS 340

Exam 2 Review

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Q1

~Code~
340

Updates

1. Exam 2 Thursday
   a. Exam review out
   b. Practice exam out
   c. 2nd chance next week Thursday

1. No HW due this week

1. MP6 Wallet due next week Tuesday

CS 340 - Computer Systems

LG: Understanding, at a high level, how we get from electricity to a website.

LG: Developing mental models of how different computer systems work so that you can understand more complex ideas if needed later.

LG: Getting experience working on unstructured, substantial coding projects.

Agenda

- Exam 2 Format
- Big Ideas Overview
- Virtual Memory
- Bit Operation/negatives
- Concurrency Programming

Find someone nearby

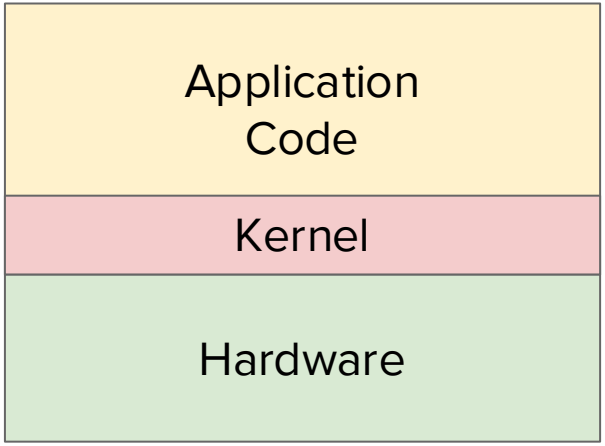
A) What is your name

B) Favorite thing you ate over spring break

Exam 2 Format

- 10pts - Conceptual MCQs.
- 5pts - UTF-8 convert question (either one of the HW questions)
- 5pts - 2’s complement conversion question (from hw)
- 10pts - 1 tricky making masks with bitwise operations variant (from hw)
- 10pts - 1 code reading race condition reasoning question
- 10pts - 1 code reading question about MP5 or MP4

Big Ideas



Code => virtual memory

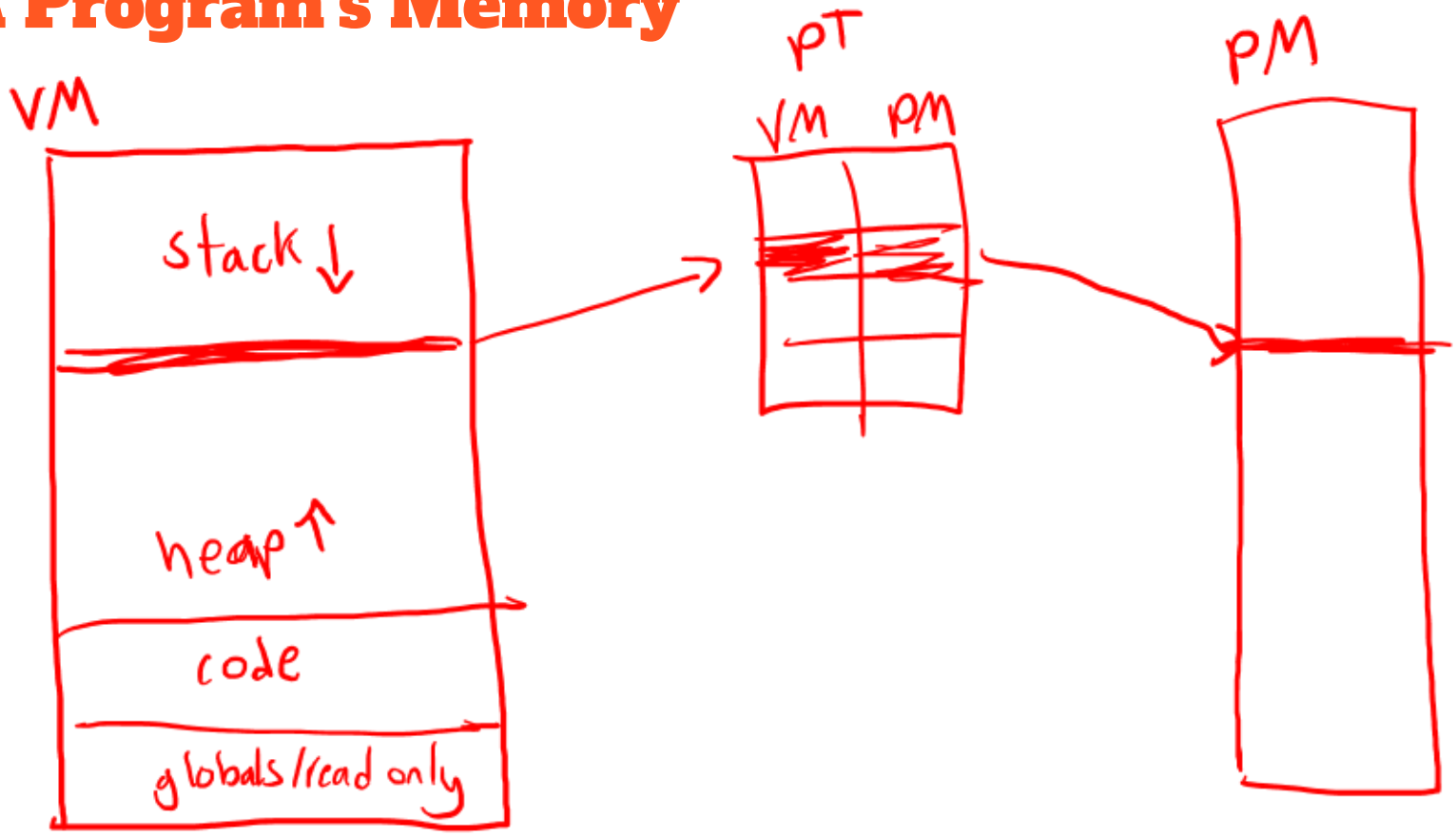
code that has special permissions.

Transistors -> gates -> logic

information storage

CPU, ALU, memory, disk

A Program's Memory



What memory segment is there an error in?

```
11 int main() {
12     int arr[5] = {1, 2, 3, 4, 5};
13     printf("%i\n", arr[5]);
14 }
```

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Q2

~Code~
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Stack

What memory segment is there an error in?

```
13 int main() {
14     int* arr = malloc(4*5);
15 }
```

heap

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Q3

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What memory segment is there an error in?

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Q4

~Code~
340

```
13 int main() {
14     char *str = "hi";
15     str[0] = 'H';
16 }
```

read only

Which are true?

True

A) To run a program that prints “hello world” bits that represent ‘h’ or 68 need to be stored somewhere physically in memory.

True

A) To run a program that prints “hello world” to your screen the OS has to manage multiple programs and devices.

True

A) To run a program that prints “hello world” to your screen your computer needs to use logic gates.

True

A) To run a program that prints “hello world” to your screen the computer needs to context switch.

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Q5

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What does a CPU/processor do?

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Q6

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340

a) It runs python, C, and other code instructions.

a) It runs 1 assembly instruction at a time using registers and the ALU.

a) It runs many instructions in parallel by utilizing the ALU.

Explain to a neighbor...

What is the difference between a thread and a process? How are they handled differently and similarly by the OS?

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Q7

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Thread

PC
registers

process

1+ threads
Page table/VM

OS schedules
at a thread level

Explain to a neighbor…

What is the difference between concurrency and parallelism?

same time

active

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Q8

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Explain to a neighbor…

What is the difference between a thread and a process? How are they handled differently and similarly by the OS?

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Q9

~Code~
340

repeat

Why Bits?

Hardware can tell between high and low voltage

1 0

→ store 1's and 0's → package into byte → represent with 2 hex digits

↑ 8 bits

Bits are Bits

- represent numerical value
- represent negative number
- represent metadata
  - UTF-8
  - Allocator
- represent a set

---

bit shifting
bit operations
pointer math

Pointer Math Review

```
int x = 9;
int *ptr = &x + 2
```

$$2\,(\underbrace{\text{sizeof}(\overset{\text{int}}{*\&x})}_{4});$$

$= 8$ bytes

What goes in the blank so "YAY" prints?

```
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    int *ptr = arr + ___;

    if(&arr[3] == ptr){
        printf("YAY\n");
    }
}
```

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Q10

~Code~
340

3

What goes in the blank so "YAY" prints?

```
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    void *ptr = arr;
    ptr = ptr + ___;

    if(&arr[2] == ptr){
        printf("YAY\n");
    }
}
```

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Q11

~Code~
340

8

What would line 20 print?

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Q12

~Code~
340

```
11 int main() {
12     int x = 4;
13     //prints 0xb7098fbc
14     printf("%#x\n", &x);
15
16     int *ptr = &x + 1;
17     void *ptr2 = (void*)ptr;
18     ptr2 = ptr2 + 1;
19
20     //prints ??
21     printf("%#x\n", ptr2);
22 }
```

+ 4 bytes

+ 1 bytes

0xb7098fbc

+ 5

0xb7098fc1

Negative Numbers Review

-7 0000 0111

flip 1111 1000

add 1111 1001

What prints?

```
11 int main() {
12     unsigned char x1 = 0b11111001;
13     char x2 = 0b11111001;
14     printf("%i and %i", x1, x2);
15 }
```

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Q13

~Code~
340

249 and -7

What prints?

0111 ? ? ? ?

1111 ? ? ? ? 0

```
11 int main() {
12     char x1 = 0x7E;
13     x1 = x1 << 1;
14     printf("%i\n", x1);
15 }
```

-4

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Q14

~Code~
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What code at line 14 would make the program print out a negative value?

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Q15

~Code~
340

```
11 int main() {
12     int x = 0x07000000;
13
14      //??
15
16     printf("%i\n", x);
17 }
```

a) x = x << 4;
b) x = x << 10;
c) x = x ^ 0x77777777;
d) x = (~0) | x;

Do the following char values make a valid UTF-8 string?

0x01 0xF7 0x88 0x84 0x98 0x1F

00... 11110 10 10 10 0...

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Q16

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Yes

Recall that UTF-8 encodes code points in bytes using the following table:

```
0xxxxxxx
110xxxxx 10xxxxxx
1110xxxx 10xxxxxx 10xxxxxx
11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
```

Printing out the 9th bit

```
12 int main() {
13     void *ptr = malloc(500);
14
15     //how would I print out the 9th bit?
16
17
18
19
20
21
22     free(ptr);
23 }
```

Chat with a neighbor, how would you fix this?

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Q17

~Code~
340

```c
int main() {
    void *ptr = malloc(500);
    void *p = ptr;
    //how would I print out the 9th bit?
    ptr = ptr + 1;
    char *ptr2 = (char*)ptr;
    unsigned char val = *ptr2;
    val = val >> 7;
    printf("%i\n", val);

    free(ptr);
}
```

p

```
main.c: In function 'main':
main.c:22:5: warning: 'free' called on pointer 'ptr' with nonzero offset
   22 |     free(ptr);
      |     ^~~~~~~~
main.c:13:17: note: returned from 'malloc'
   13 |     void *ptr = malloc(500);
      |                 ^~~~~~~~~~
0
munmap_chunk(): invalid pointer
```

Manual Pages for malloc/free/calloc/realloc

DESCRIPTION

The **malloc**() function allocates *size* bytes and returns a pointer to the allocated memory. *The memory is not initialized*. If *size* is 0, then **malloc**() returns either NULL, or a unique pointer value that can later be successfully passed to **free**().

The **free**() function frees the memory space pointed to by *ptr*, which must have been returned by a previous call to **malloc**(), **calloc**(), or **realloc**(). Otherwise, or if *free(ptr)* has already been called before, undefined behavior occurs. If *ptr* is NULL, no operation is performed.

Chat with a neighbor, what is the definition of a race condition?

- ☐ threads
- ☐ shared resources
- ☐ undetermined outcome

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Q18

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This code sometimes prints 1 and sometimes 0, is there a race condition?

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Q19

~Code~
340

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    void *ptr_front = malloc(500);

    //how would I print out the 9th bit?
    void* ptr = ptr + 1;
    char *ptr2 = (char*)ptr;
    unsigned char val = *ptr2;
    val = val >> 7;
    printf("%i\n", val);

    free(ptr_front);
}
```

No

no threads

```c
char str[6] = {'h','e','l','l','o','\n'};

void *comp(void *mem){
    char **replace = (char**)mem;
    char *ptr = str;
    while(*ptr != '\n'){
        if(*ptr == replace[0][0]){
            *ptr = replace[1][0];
        }
        ptr++;
    }
}
int main() {
    char* replace0[2] = {"y", "m"};
    char* replace1[2] = {"o", "0"};
    char* replace2[2] = {"h", "y"};
    pthread_t threads[3];
    pthread_create(&threads[0], NULL, comp, replace0);
    pthread_create(&threads[1], NULL, comp, replace1);
    pthread_create(&threads[2], NULL, comp, replace2);
    for(int i = 0; i < 3; i++){
        pthread_join(threads[i], NULL);
    }
    printf("%s\n", str);
}
```

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Q20

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What are two possible outputs?

yell0

mell0

Synchronization Summary

Errors -

race condition

deadlock

Synchronization tools -

Mutex

reader/writer

CV

Wallet

Static Condition Variable Example

```c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>

pthread_mutex_t my_mutex = PTHREAD_MUTEX_INITIALIZER;
pthread_cond_t my_cv = PTHREAD_COND_INITIALIZER;

int shared_int = 0;
int pred = 0;

void *shared_function(void* arg){
    pthread_mutex_lock(&my_mutex);
    while(pred == 0){
        pthread_cond_wait(&my_cv, &my_mutex);
    }
    shared_int = pred * 5;
    pred++;
    pthread_cond_broadcast(&my_cv);
    pthread_mutex_unlock(&my_mutex);
}

//thread creation code and main function, changes pred to 1 eventually
```

```c
void *first_computation(void *arg){
    pthread_mutex_lock(&my_mutex);
    printf("FIRST %i\n", shared_int);
    pred = 1;
    pthread_cond_broadcast(&my_cv);
    pthread_mutex_unlock(&my_mutex);
}
```